Manual Calculation.

Exp 2:

1. Program A:

X = A7H = 1010 0111

Y = 3CH = 0011 1100

W = 1234H

AND

X	1010	0111
Y	0011	1100
	0010	0100

XANDY = 24H

OR

X = 1010 0111

Y = 0011 1100

1011 1111

X OR Y = BFH

XOR:

X = 1010 0111

Y = 0011 1100

1001 1011

XXOR Y = 9B H

Upper-Nipple masking

X = 1010 0111

mask = 1111 0000

1010 0000

UNM = A0H

One's complement:

X = A7H = 1010 0111

X' = 0101 1000

Result = 58H

2's complement

X' = 0101 1000

1

0101 1001

Result = 59H

1's complement of 16bit W

W = 1234 H

W = 0001 0010 0011 0100

W' = 1101 1101 1100 1011

Result: EDCBH

2's complement of 16-bit W.

EDCBH
1H
EDCCH

Flag calculation after AND
Result: 24H, ⇒ 0010 0100
S=1, Z=0, AC=1, P=1, CY=0.

2 Program B:

Shift & Rotate operation.

D = CAH = 1100 1010
W = 9234 H

Set bits 7 2 0; Mask 85H.

D = 1100 1010
Mask = 1000 0001
1100 1011

clear bits 3, 2, 1

Mask = 75H

D = 1100 1010
Mask = 1111 1010
1100 0000 result: C0H.

Toggle Lower Nibble:

Mask = 0FH

D = 1100 1010
Mask = 0000 1111
1100 0101 result = C5H

Test Bit 6:

Mask: 40H

D = 1100 1010
Mask = 0100 0000
0100 0000 ∴ Result = 40H

Rotate left (RLC)

D = 1100 1010

ori = 1100 1010

Rot = 1001 0101

95H ⇒ CY = 1

Rotate right (RRC)

ori = 1100 1010

Rot = 0110 0101

65H CY = 0.

Logical shift left.

ori = 1100 1010

shift : 1001 0100

94H CY = 1

Arithmetic shift right:

ori = 1100 1000

shift = 1110 0101

E5H ; CY = 0.

Count of 1's

D = 1100 1010

No. of 1's = 4.

16-bit shift left